6061 Introduction to Deep Learning

Exercise 2

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This exercise sheet will not be graded and is provided solely as practice.

1. **Linear Algebra** From Mathematics for Machine Learning Chapter 2:
   (a) Exercise 2.4

2. **Derivatives** Based on the Deep Learning course from University of Tübingen.
   (a) For $a, x \in \mathbb{R}^n$, show that $\nabla_x a^T x = a^T$.
   (b) Let $\mathbf{f}(\mathbf{x}) = (\exp(x_1)\sin(x_3), \sqrt{x_2 + x_3}), \mathbf{f} : \mathbb{R}^3 \to \mathbb{R}^2$. Calculate the matrix of partial derivatives $\frac{\partial \mathbf{f}}{\partial \mathbf{x}}(\mathbf{x})$.
   (c) Let $a : \mathbb{R} \to \mathbb{R}$ and define $\mathbf{f}(\mathbf{x}) = (a(x_1), \ldots, a(x_n)), \mathbf{f} : \mathbb{R}^n \to \mathbb{R}^n$. Calculate the matrix of partial derivatives $\frac{\partial \mathbf{f}}{\partial \mathbf{x}}(\mathbf{x})$. What property does the resulting matrix have?
   (d) Let $\sigma(x) = \frac{1}{1+\exp(-x)}, \sigma : \mathbb{R} \to \mathbb{R}$. Show that $\sigma'(x) = \sigma(x)(1 - \sigma(x))$.
   (e) Calculate the derivative $\tanh'(x)$ with $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}, \tanh : \mathbb{R} \to \mathbb{R}$.

3. **Probability**
   (a) Show that $E(aX) = aE(X)$.
   (b) Show that $E(a) = a$.
   (c) Show that $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$.
   (d) Is $p_X(x) = 3x^2 - e^x$ for $2 \leq x \leq 4$ a valid probability density function? If not, how could the function be modified for it to be a valid PDF?